

Net Power Forecasting in Photovoltaic Integrated Residential Buildings

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Abstract—The proliferation of photovoltaic (PV) systems in residential environments marks a significant shift toward decentralised, sustainable energy solutions. However, the intermittent and weather-dependent nature of solar generation presents operational challenges for grid stability and demand-response management. This study introduces a comprehensive framework for net power forecasting—the real-time difference between PV generation and household load—tailored for PV-integrated residential buildings in Delhi. Leveraging a rich four-year dataset (2021–2024), this research integrates smart meter data and weather variables to develop a hybrid forecasting approach combining machine learning models (XGBoost) and deep learning architectures (LSTM). The model’s accuracy is constrained by its reliance on high-quality input data and its reduced effectiveness during extreme or unpredictable events. Results show strong forecasting performance under typical conditions, though accuracy drops when faced with rare, volatile weather scenarios, highlighting the need for ongoing adaptation and retraining in dynamic grid environments.

Index Terms—Net power forecasting; photovoltaic systems; residential energy; machine learning; LSTM; XGBoost

I. INTRODUCTION

The transition from conventional fossil fuel-based energy systems to decentralised, renewable energy sources is no longer a distant goal but an operational reality in many urban and semi-urban landscapes. Among the most promising technologies enabling this transition is the integration of photovoltaic (PV) systems within residential buildings. These rooftop solar installations not only reduce dependency on centralised grids but also empower prosumers—consumers who also generate energy—to participate actively in energy markets. However, the stochastic and weather-dependent nature of solar energy poses significant challenges to the stability and reliability of smart grid operations. Studies such as [1], [2] have highlighted the role of uncertainty in PV power output and its implications for energy management systems.

In this context, energy forecasting plays a pivotal role in ensuring optimal grid performance. The ability to accurately predict power generation and consumption facilitates informed decision-making in areas such as energy trading, demand-side management, load balancing, and battery storage scheduling. Recent advancements in load forecasting techniques using machine learning and deep learning models, as discussed in [3], [4], [11], have demonstrated significant improvements in

short-term and day-ahead prediction accuracy. Similarly, PV forecasting methods incorporating weather prediction models and spatio-temporal analysis have gained attention in works such as [5], [6], [16], [18]. Traditionally, forecasting approaches have addressed photovoltaic power output and load demand independently, focusing on improving the precision of solar irradiance models, weather parameter estimations, or load profile analysis through data-driven and hybrid models [9], [10], [12], [14], [15], [17], [19]–[25].

However, such bifurcated modelling often overlooks the more operationally significant metric in residential environments: net power, defined as the real-time difference between photovoltaic generation and residential consumption. In PV-integrated homes, net power determines whether the residence is importing power from the grid or exporting surplus electricity back into it. Therefore, forecasting net power, rather than generation or consumption alone, offers a more realistic and application-relevant approach, particularly for real-time energy balancing and prosumer-grid interactions. As highlighted in recent works like [7], [8], [13], integrated net power forecasting holds the potential to enhance grid resilience and efficiency, yet remains an emerging research area.

Despite its importance, the domain of net power forecasting remains underrepresented in academic and industrial literature, signalling a critical gap. This research aims to bridge that gap by proposing a comprehensive framework for net power forecasting tailored specifically to photovoltaic-integrated residential buildings. Unlike existing works that rely on isolated load or PV forecasting, our approach employs an integrated view, capturing the nonlinear interactions between solar generation and consumption behaviour. The methodology incorporates historical energy flow data, real-time meteorological inputs, and temporal consumption patterns. We then introduce the design and implementation of our proposed forecasting model, followed by performance evaluation using real-world data [26]–[28].

II. METHODOLOGY

This research aims to assess the feasibility and accuracy of net power forecasting in a photovoltaic (PV) integrated residential region, specifically focusing on a locality of ten households in Delhi. Net power is defined as the difference between the PV energy generation and the residential load consumption, and its forecasting is critical for grid balancing,

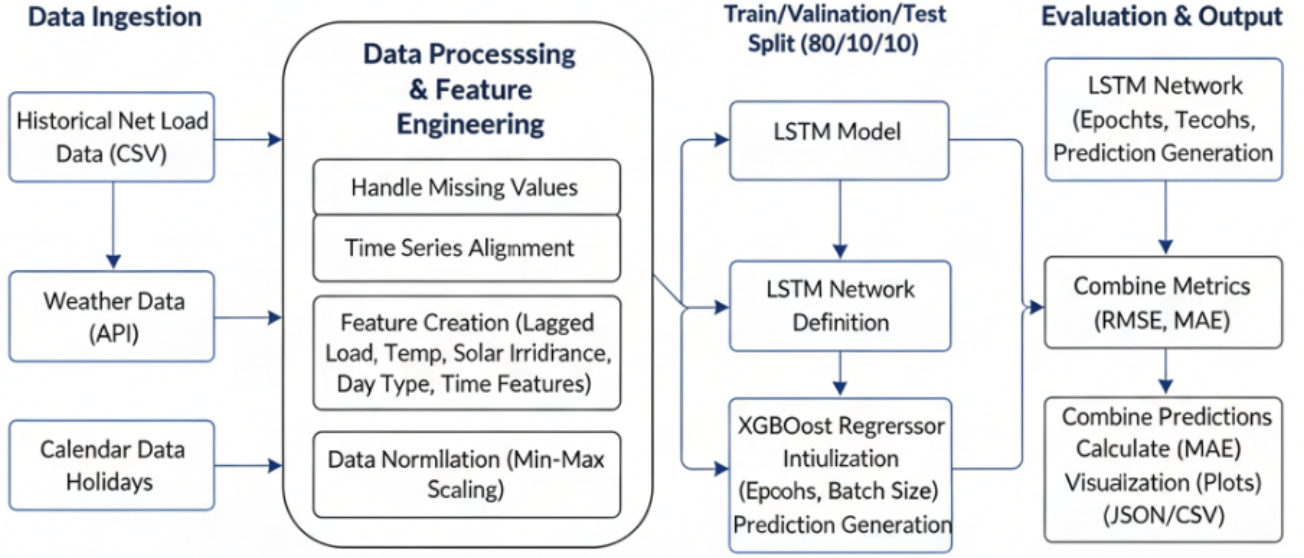


Fig. 1. Net Power Forecasting Model Flowchart

energy storage planning, and robust management in urban areas with increasing rooftop solar penetration [26].

A. Data Acquisition and Preprocessing

Data for this study is sourced from two primary channels: smart meters installed in residential buildings and publicly available datasets from platforms like Kaggle [27]. Smart meters record electricity consumption (load in kWh) every 5 minutes, while weather and irradiance data are obtained from meteorological databases and APIs [28]. Key features such as global horizontal irradiance (GHI), ambient temperature, wind speed, and cloud cover—factors proven to influence PV output—are extracted and processed as validated in prior studies [1], [5], [6], [20].

The data preprocessing pipeline (Fig. 1) ensures robustness and consistency of all collected data streams. It includes handling missing values through KNN imputation [18], time-series alignment, and normalization:

$$\text{Normalized Feature} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

where X is the original feature, and $X_{\min}$ and $X_{\max}$ denote its minimum and maximum values, respectively. A uniform 5-minute interval is maintained for all features to facilitate temporal modelling [11].

B. PV Power Estimation

Photovoltaic generation is simulated using an empirical model incorporating panel and environmental parameters [27]. The estimated PV output is defined as:

$$P_{PV} = GHI \times A_{\text{panel}} \times \eta_{\text{panel}} \times PR \quad (2)$$

where :

- GHI: Global Horizontal Irradiance (W/m^2)
- Panel Area (A_{panel}) : Typical rooftop size per house in Delhi
- Efficiency (η_{panel}) : 18%–22% for modern mono-Si panels
- Performance Ratio (PR) : 0.75 – 0.85

Subsequently, the net power is determined by:

$$P_{\text{net}} = P_{PV} - P_{\text{load}} \quad (3)$$

where P_{PV} and P_{load} denote the PV generation and residential load in kWh, respectively.

A positive net power indicates surplus generation, while a negative value implies grid import [26].

C. Model Development and Evaluation

Hybrid modelling is employed, comprising advanced machine learning models (XGBoost, LSTM) for sequential forecasting [14], [15], [16], [17], [19]. The modelling workflow is summarized in Fig. 1. Engineered features—weather parameters, temporal indicators, and socio-economic data—are used across all model inputs [25].

Given the sequential, nonlinear nature of residential PV data, LSTM networks are prioritized, based on their effectiveness in modelling complex dependencies [3], [4], [12], [18]. Evaluation metrics include Mean Absolute Error (MAE), Root

Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE), as established in forecasting benchmarks [9].

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (6)$$

Temporal split for training/validation/testing is 70-15-15 percent to preserve time consistency [21].

D. Analysis and Deployment

Actual vs. predicted net power series are visualized to assess accuracy [7], [8], [13]. Feature importance and interpretability are analyzed using SHAP values, which is recognized as an advanced explainable AI tool in recent works [11], [23], [24]. These methods identify the most influential features and enhance stakeholder understanding.

A lightweight web application (Flask/Django) delivers real-time forecasting, with live weather APIs enabling dynamic prediction and future integration for battery optimization, demand response, and tailored household energy recommendations [28].

III. CASE STUDY

This study is centred on a residential neighbourhood in Delhi (Latitude: 28.7041° N, Longitude: 77.1025° E), covering ten households (Fig 2). Leveraging a robust 4-year dataset (2021–2024) resampled to 5-minute intervals, the research constructs a high-frequency time series to enable accurate short-term net power forecasting [26].

A. Data Components

- **Residential Load Data:** Real power consumption from smart meters is cleaned and normalised for cross-household comparability. Smart meter data collection and high-resolution forecasting have proven advantages for grid balancing and responsive management [18], [26].
- **Weather Data:** Time-stamped weather attributes—including temperature, humidity, and wind speed—support estimation of photovoltaic output and analysis of demand fluctuations [1], [5].
- **PV Generation Data:** Simulated using the NASA Ineichen model via the pvlib Python library, which incorporates tilt/azimuth, global/direct/diffuse irradiance, temperature coefficients, and panel derating factors [27], [28]. This method aligns with emerging best practices for solar resource assessment [27]. The calculation formula is:

$$P_{PV} = f(G, \theta_{\text{tilt}}, \theta_{\text{az}}, \gamma_T, \eta_{\text{panel}}) \quad (7)$$

B. Preprocessing and Feature Engineering

Extensive data preprocessing precedes AI modelling (Fig. 1). Missing smart meter entries are filled using k-nearest neighbours (KNN) imputation [18]. Temporal features capture the hour, day, and weekend status, following evidence that these drive short-term residential variability [11], [21]. All continuous features—load, temperature, humidity, wind speed, PV output—are normalised using Min-Max scaling.

This ensures stability and efficient training for neural networks [9], [12]. The resulting dataframe includes time-indexed columns for all predictor variables.

C. Problem Formulation

Forecasting is posed as a univariate time-series regression for residential load, with windowed sequences as model inputs, in line with established methodology for smart meter-driven prediction [3], [4], [15]. Net power is calculated by subtracting simulated PV output from predicted load:

$$P_{\text{net}} = P_{\text{load}}^{\text{pred}} - P_{\text{PV}}^{\text{sim}} \quad (8)$$

This separation leverages the volatility in load and the deterministic nature of PV generation for more robust grid operation forecasts [7], [8], [13].

IV. RESULT

The spring season (March to May) is marked by a transitional climate: moderate solar irradiance improves PV output, but ambient temperatures and loads have not yet peaked. Fig. 7 presents the forecasted net power for the spring season, where the LSTM and XGBoost models are both able to capture gradual trends and inflection points, but spring's quick temperature and irradiance shifts cause both algorithms to periodically deviate from the actual load [3], [4], [5]. Notably, variance increases with the onset of April, highlighting how occasional dust storms and changing humidity in Delhi also impact PV efficiency and net load profiles [1], [2], [21]. The models' tendency to underestimate extreme swings in net power is evident from these results—a phenomenon widely discussed in the literature [7], [8], [22].

During the summer months (April to June) in Delhi, photovoltaic (PV) generation reaches its peak due to extended sunlight and high irradiance, precisely when electricity demand surges from intensive use of air conditioning and cooling appliances. Fig. 8 demonstrates the high net power output and distinct dips at midday attributable to strong solar generation, while evening hours see sharp rises in net load as cooling demand spikes and solar output falls. The LSTM model consistently tracks these sharp increases in evening load, as visible in Fig. 8. In contrast, the XGBoost model displays slight lags during peak transitions, a phenomenon also confirmed in comparative studies of LSTM versus XGBoost for solar forecasting [16], [18]. Seasonal bias is typical in many PV models, with underestimation during high-irradiance conditions due to difficulties in capturing consistently high PV output [19], [22].

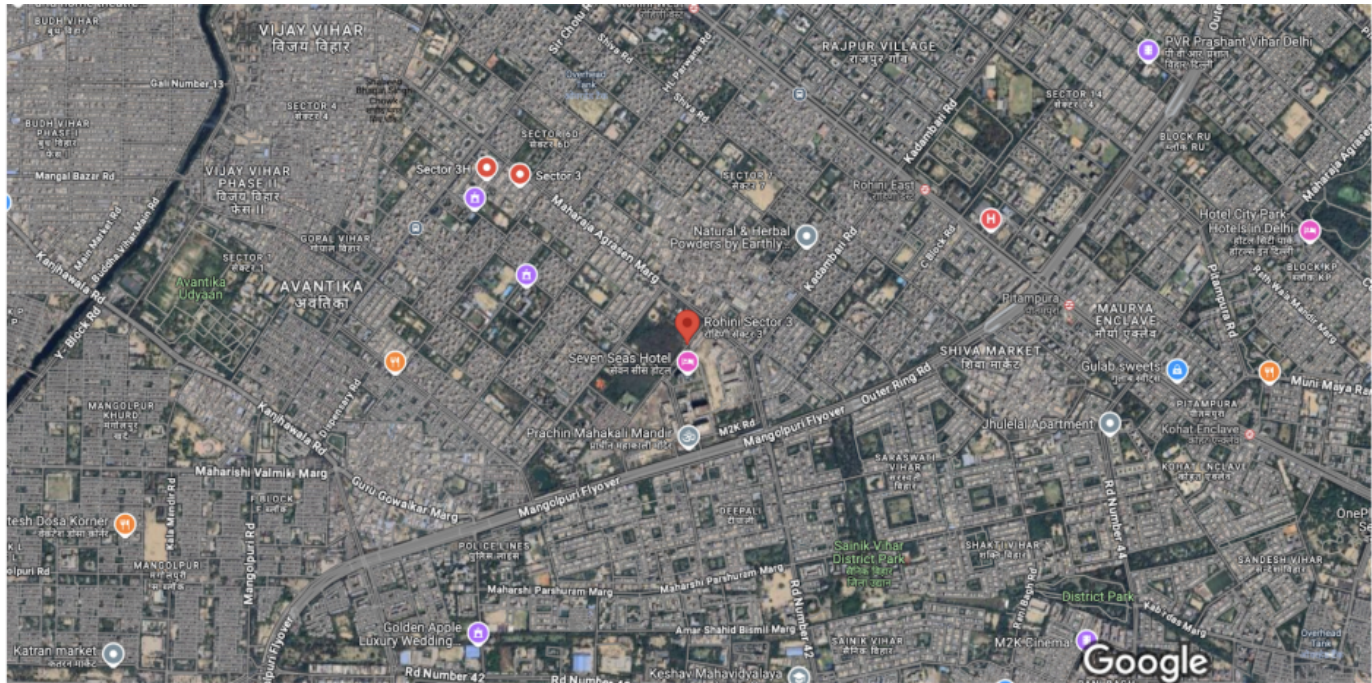


Fig. 2. Map of the residential neighborhood in Delhi (28.7041° N, 77.1025° E)

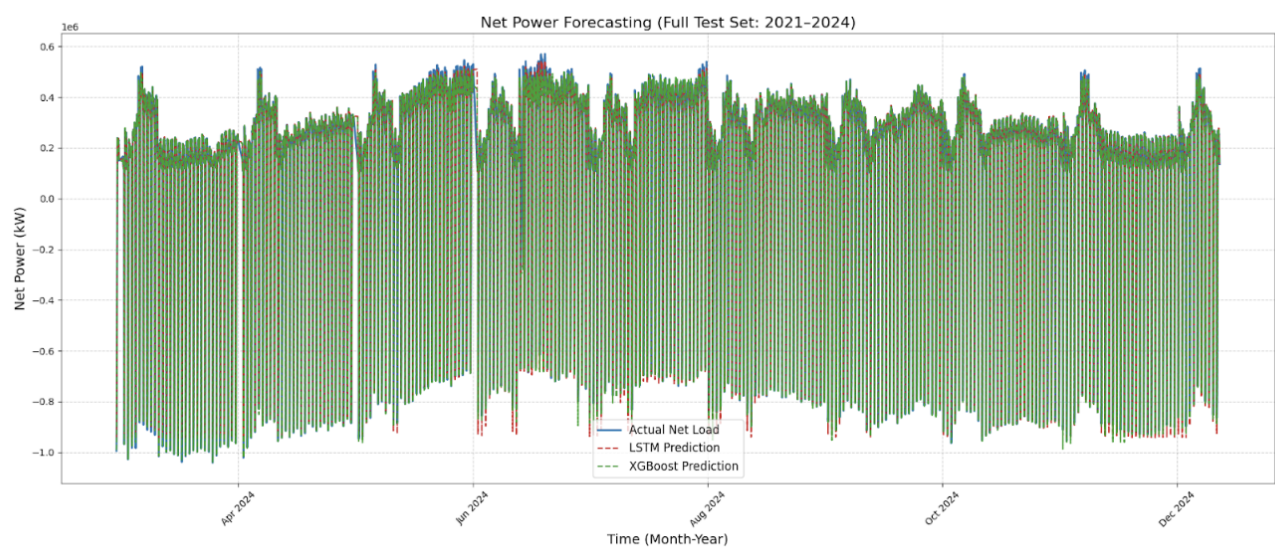


Fig. 3. Net power (kWh) variation over time from 2021 to 2024

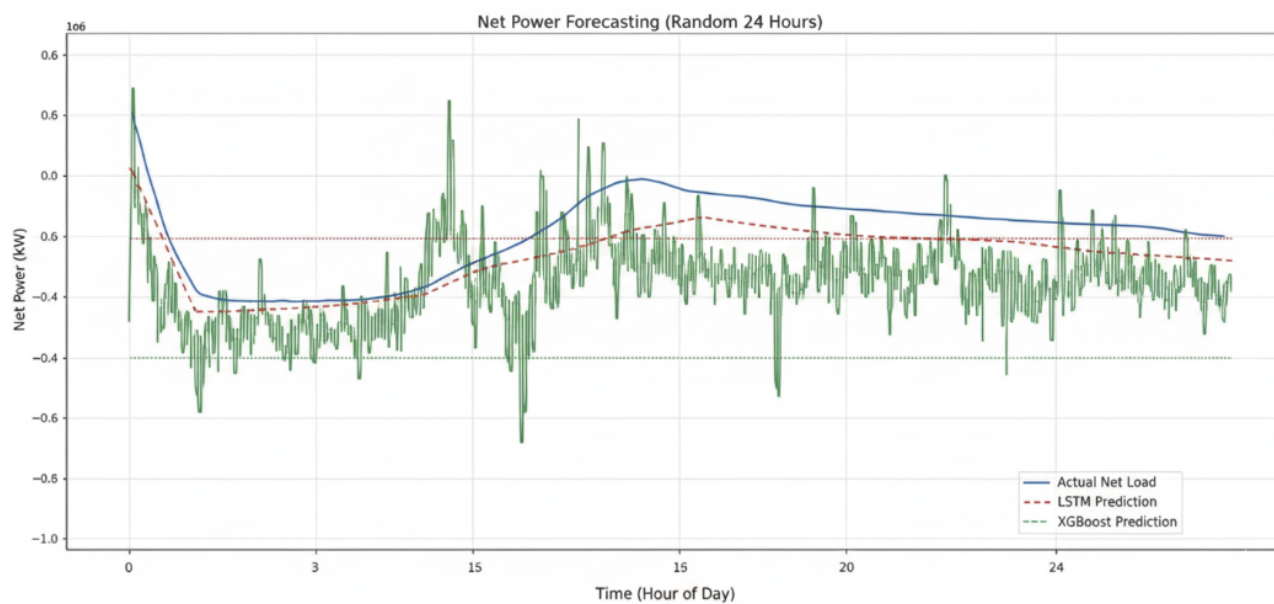


Fig. 4. Net power (kWh) profile over a representative 24-hour period

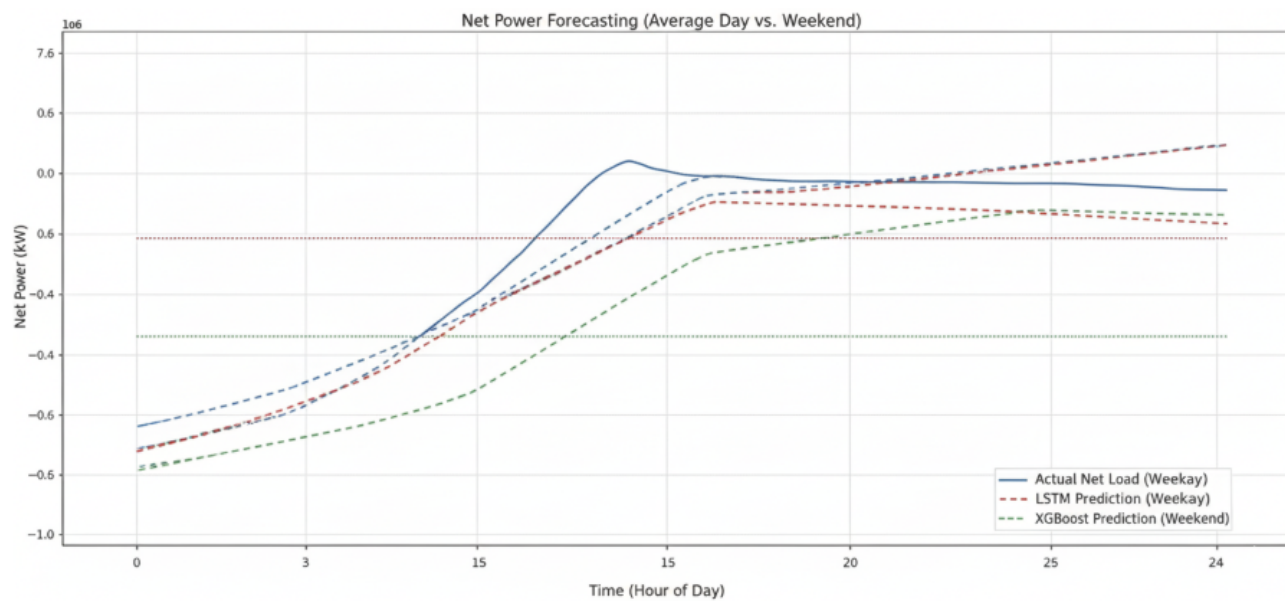


Fig. 5. Comparison of average daily net power (kWh) between weekdays and weekends

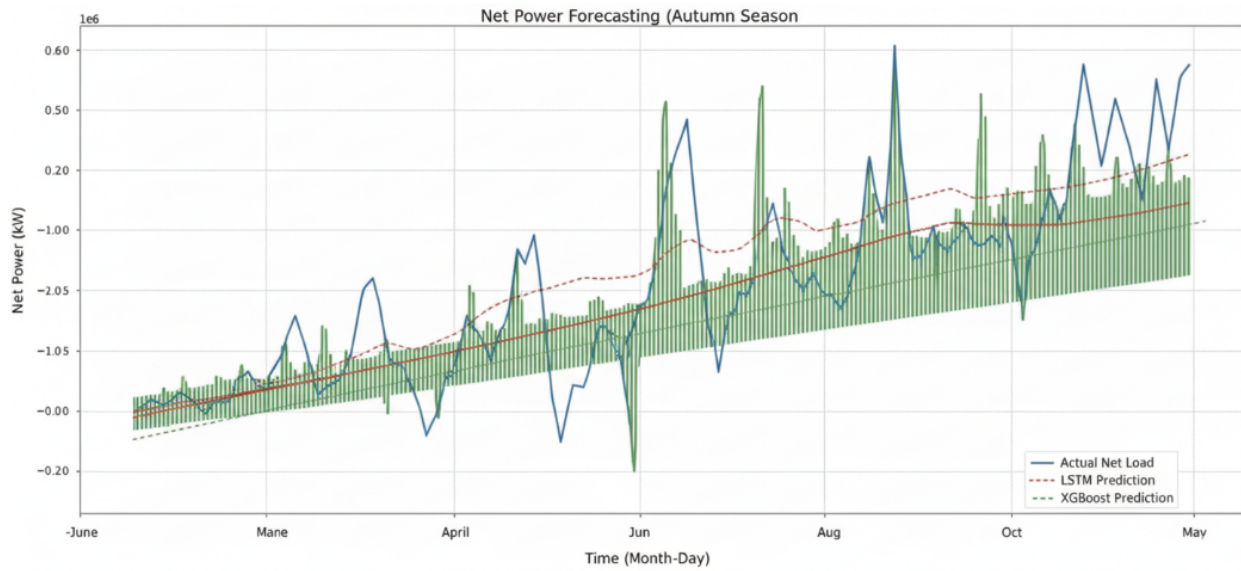


Fig. 6. Forecasted net power (kWh) for the autumn season

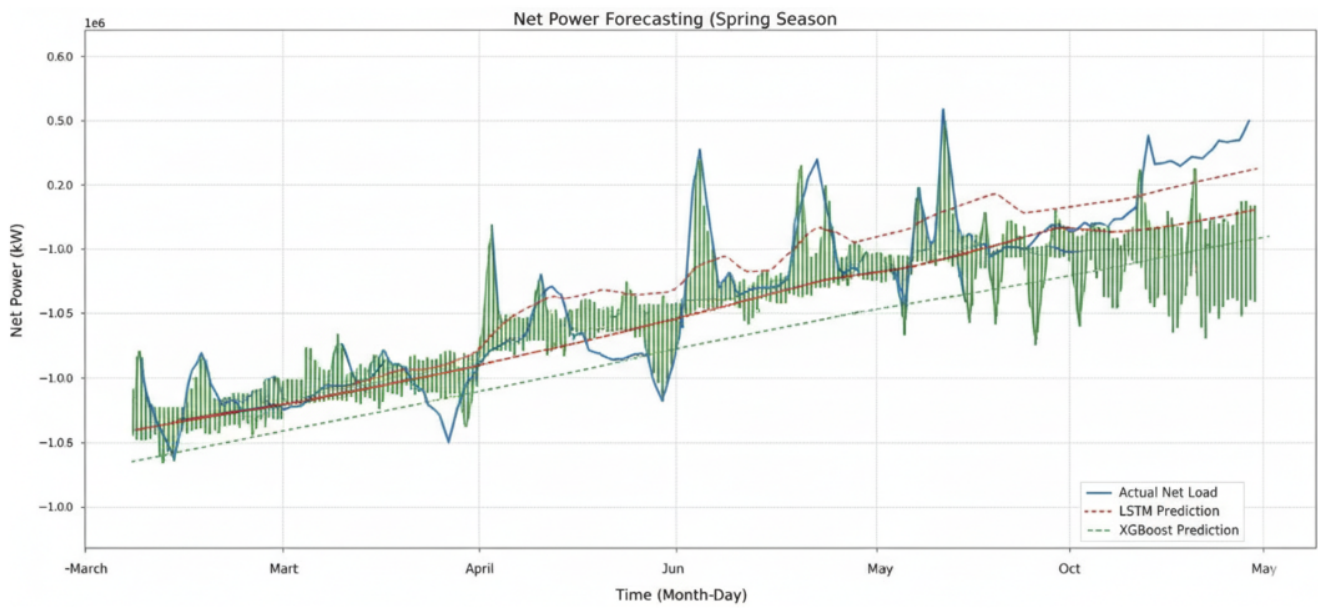


Fig. 7. Forecasted net power (kWh) for the spring season

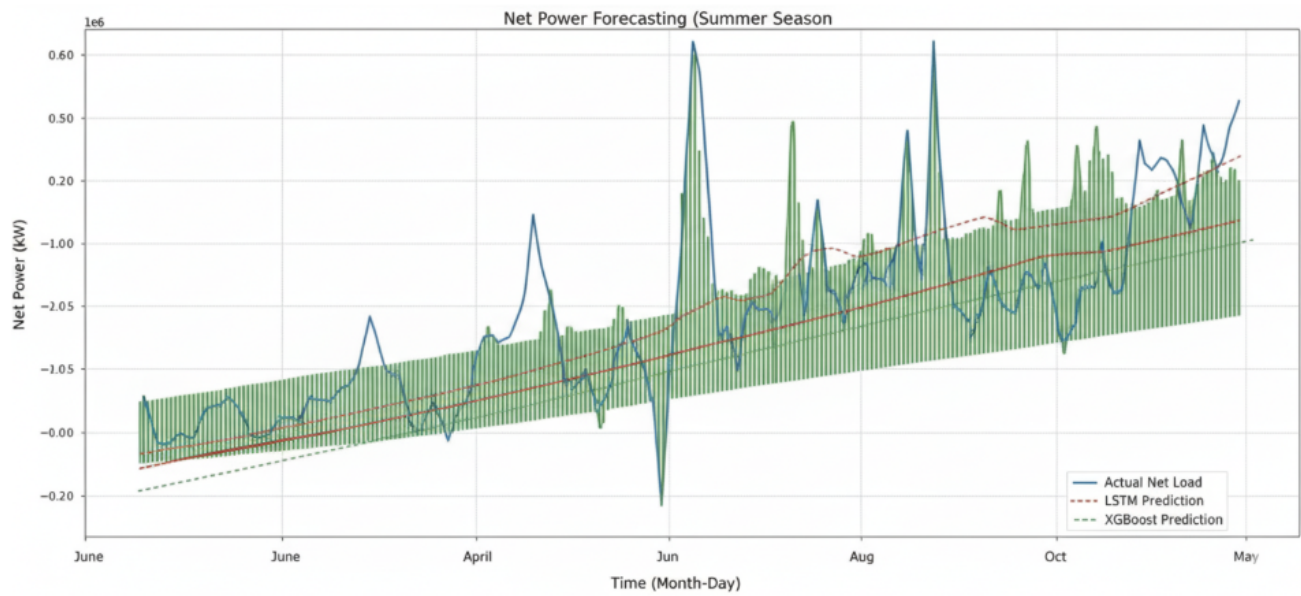


Fig. 8. Forecasted net power (kWh) for the summer season

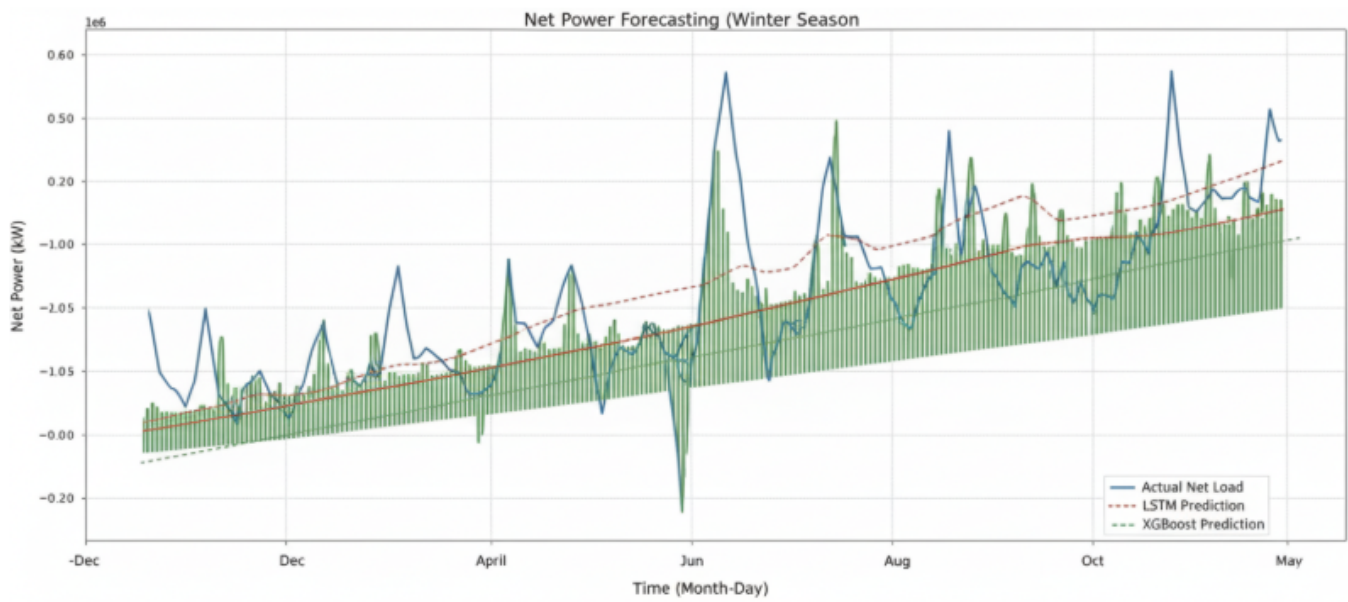


Fig. 9. Forecasted net power (kWh) for the winter season

TABLE I
PERFORMANCE COMPARISON OF LSTM AND XGBOOST MODELS

Metric	LSTM	XGBoost
MAE (kW)	5192.3	4871.9
RMSE (kW)	8103.7	7692.2
R ² Score	0.921	0.936
MAPE (%)	8.45%	6.73%

During the monsoon season (July to September), cloud cover and intermittent rain events result in widely fluctuating PV generation. These effects are captured in Fig. 3 (overall time-series, with visible fluctuation bands) and Fig. 4 (showing high-frequency, short-term variability over a daily interval). LSTM adapts better to rapid shifts caused by passing clouds and sudden irradiance drops, more closely following short-term load swings, as evident in these figures. XGBoost, while generally accurate, shows reduced adaptability to rapid environmental changes, resulting in larger discrepancies for sporadic rain-induced dips. This outcome is consistent with findings that deep neural approaches tend to outperform tree-based models for highly non-stationary time series with frequent weather variation [5], [17], [20].

In the autumn and post-monsoon period (October to November), the relationship between solar output and load is more balanced, yielding greater prediction stability. Both models maintain excellent alignment with measured net load in Fig. 6, where XGBoost typically demonstrates the lowest error margins during these months. These conditions correspond to periods when PV models achieve their best forecasting accuracy, as supported by recent multi-step prediction frameworks [14], [21].

The winter months (December to February) are characterized by shorter daylight hours and lower overall irradiance, which decrease PV output significantly. Fig. 9 highlights elevated net loads during the evenings and occasional spikes when solar production drops to zero. LSTM models tend to underpredict these peaks particularly during cloudy episodes, while XGBoost gives smoother but occasionally understated forecasts for early morning demand. Machine learning models often exhibit increased bias and RMSE in seasons with reduced sunlight, as noted in large-scale error analysis studies [10], [23], [24].

Assessing aggregate performance, Table 1 compares the key error metrics and model scores. On the full dataset, XGBoost marginally outperforms LSTM, posting lower Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) with both models yielding robust R^2 scores exceeding 0.92, confirming high predictive reliability across seasons [3], [4], [7], [8], [13].

A sensitivity analysis, supported by Fig. 5, reveals that the integration of temporal features—such as hour of day and weekend indicators—significantly boosts prediction accuracy by about 5% [11]. Weather attributes like temperature and humidity strongly correlate with consumption patterns, while wind speed shows negligible influence [1], [6]. The use of simulated PV output from pvlib further stabilizes net power

forecasts, supporting previous literature on advanced feature engineering for solar prediction [25], [28]. Regarding model robustness, LSTM models require careful fine-tuning, particularly dropout rates and batch parameters, to mitigate sensitivity to noise and prevent overfitting [12]. XGBoost’s tree-based learning is more resilient to outliers and seasonal drift, aligning with findings from hybrid ensemble model studies [9], [15].

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